

From archive to experience: Reimagining digital archives as immersive knowledge spaces

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Abstract

In the contemporary practice of documentation and preservation of cultural heritage, the creation of larger datasets to generate high-resolution 3D representations has already become widespread (e.g., Salagean-Mohora, Anghel & Frigura-Iliasa, 2023). To enrich the created digital representations, collection of metadata to link to the models is common practice. These digital archives can then be processed for presentation and dissemination (Baillie, 2024) using a wide variety of strategies, such as immersive media exhibitions, extended reality applications and web-based formats. As such, knowledge dissemination to the public, and the underlying archival documentation, are usually understood as two sequential, siloed stages.

Where current practice treats integrating content and narrative only at the presentation layer, while leaving the archival structure untouched, this research collapses them, so that the archive's spatial organisation becomes the interpretive layer, serving both preservation and meaning-making. Rather than treating archives as repositories of isolated assets that risk becoming "data bunkers", in which metadata-rich 3D objects lack the context necessary for personal interpretation (Amineh & Asl, 2015), the research conceptualises them as navigable narrative spaces. Arranged into a relational room programme, virtual spaces and digital content combine spatial organisation with digital storytelling to support personalised meaning making, interaction and knowledge retention (Abderrahim & Gutiérrez-Colón Plana, 2021).

The employed method draws on photogrammetry scans, CAD plans, 3D reconstructions, audiovisual material and descriptive narratives to create expandable digital environments. It is tested through two deliberately unique case studies that hold this relational principle constant while varying the medium. The first, shown in Figure 1, examines Himalayan temples and religious heritage, where a web-based archive was reimagined as a "research blog." Within a single medium, a dense

network of associative links between entries and sub-pages, together with the first-ever 3D architectural glossary, fuses visual, terminological and relational knowledge into one navigable whole, without immersive technology. The second, visible in Figure 2 and focusing on a Slovene museum collection, instead unifies two heterogeneous paradigms across media in a hybrid web–Virtual Reality (VR) workflow that also documents its own development to preserve the history of the media archive (Arden, 2022). Website and VR form a single synchronised structure, each expanding directly into the other’s content. Together the two case studies triangulate a shared structural logic, within a medium and across media, locating the contribution in the archive’s organisational concept rather than any single technical solution or display tool.

Results indicate that prioritising information over simple inventory lists, while integrating user interaction into the design process, makes archives more accessible, digestible and immersive. The Himalayan case shows how associative linking between textual, visual and spatial content facilitates navigation through increasingly large and complex 2D and 3D datasets. The Slovene hybrid model offers a blueprint for documenting immersive media experiences and a scalable approach to access them across hardware environments.

Ultimately, this paper contributes to the field by proposing standardisable principles for incorporating metadata, structuring narratives and enabling cross-platform dissemination of 3D data, bridging the gap between expert-level documentation and accessible, personalised meaning making.

CHAADA

Chamba architecture and art digital archive

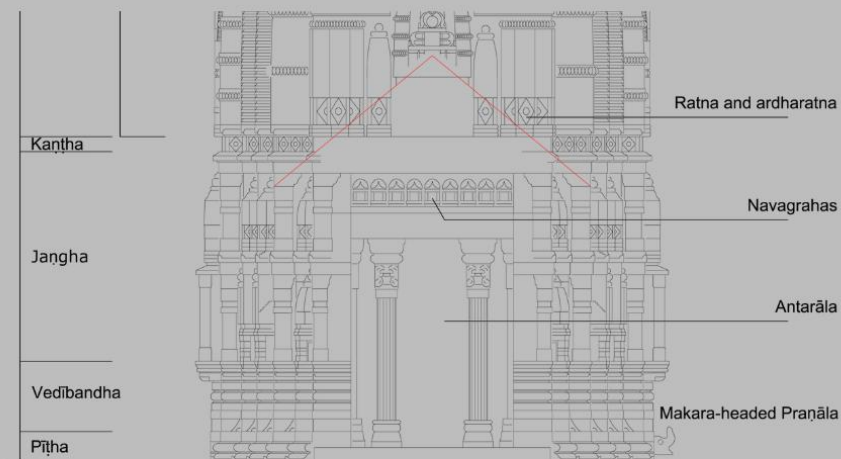
[Home](#) [Articles](#) [Gallery](#) [Glossary](#) [Regions](#) [Workflow](#)

What we do is what we blog:

The project aims at using digital technology – sometimes in a very simple and basic manner – for the documentation of cultural heritage in the region of the Western Himalaya and its foothills, a region wherein the former kingdom of Chamba was a major political and cultural factor for more than a millennium. We further aim at using digital technology to develop specific methods of representation for the different kinds of objects and data collected in our surveys. These methods include – but are not limited to – modelling of 3D-objects, 3D-printing, diagrams, plans, turntable movies, films as well as AR (augmented reality) and VR (virtual reality) environments. The goal is a contribution to digital cultural heritage preservation – a creative approach and an invitation to learn more about the cultural history of the region where the plains of Northern India border the Himalayas.



04.03.2024 – New category added – Glossary



© Gerald Kozicz – Plan of the Gaushankar Temple of Chamba Town, Chamba, India, 2024

Summary:

As on of our biggest updates to our blog so far, this week we are proud to present a new category: Glossary. It will, as the name suggests, act as a digital glossary for all the different terminology used in this blog and will feature imagery, descriptions, plans as well as a 3D model to make the contents of this blog as easy to understand as possible.

The new category can be accessed either via the site menu at the top of the page or directly [HERE](#).

Fig. 1 Main page of the CHAADA research blog.

An immersive journey to (re)trace collection and curation of Asian cultural heritage in the West

What it's all about:

The project aims to digitise and document the Slovene Ethnographic Museum's Skušek collection using digital technology. A structure-from-motion approach called "Image-based photogrammetry" thereby plays an important role, as it allows for the creation of precise digital 3D models, which basically act as a digital twin to the original object. These twins, in combination with manual, more conventional documentation methods, such as sketches, annotated drawings, photographs and videos as well as descriptive texts will build the baseline of our digitisation effort, while also serving as both documentation and preservation of the collection's current condition.

An additional goal is the reconstruction of the collection's early history and the motivations of the Skušeks, both Ivan's and Tsuneko's. Understanding the broader picture will be paramount for gaining a better understanding of their selection process when displaying, exhibiting but also using such wide variety of objects and art pieces as part of their everyday life.

Additionally, we aim to use digital technology to develop new means of presenting our research findings. By connecting our gained insights with the created digital twins, thus transforming them from simple representations into small databases, knowledge models, so to speak, we aim to make our findings and results universally accessible. This will be done by using the virtual space as a stage to create a digital museum. It will house the objects not only as virtual representations of their forms, but also as media conveying their intangible properties, that is historical and art historical, or – as in the case of Buddhist sculptures – also their religious significance. The virtual exhibition space thereby becomes a spatial field of experience which reaches beyond the pure digitisation of forms and is at the core of our dissemination strategy.



Fig. 2 Sub-page of the immersive experience.

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